THE BLASCHKE CONDITION BELOW TWO

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> Do the zero sets of functions in $\mathscr{H}^p$ for $p < 2$ satisfy the Blaschke condition in the half-plane $\operatorname{Re} s > 1/2$?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 2.3.

Answered: no. Evidence: Sketch.

No. One function belongs to every $\mathscr{H}^p$, $p < 2$, and has a non-Blaschke zero set in a bounded window.

Proof. Let $A = 64$, $a_k = A^k$, $N_k = \lceil \exp(\exp(A^{8k})) \rceil$, $L_k = \log N_k$, and

$$F(s) = \sum_{k \geq 1} \frac{a_k}{\sqrt{N_k}} P_{N_k}(s), \qquad P_N(s) = \sum_{n \leq N} n^{-s}.$$

Harper's *Moments of random multiplicative functions, I: Low moments, better than square-root cancellation, and critical multiplicative chaos*, Theorem 1, doi:10.1017/fmp.2019.7, gives $\|P_N\|_{\mathscr{H}^p} \leq C_p \sqrt{N}/(\log\log N)^{1/4}$ for each fixed $p < 2$. Hence the kth block has norm at most $C_p A^{-k}$; summability (or the p-triangle inequality for $p < 1$) gives $F \in \mathscr{H}^p$ for every $p < 2$. The positive coefficient series converges absolutely for every $\operatorname{Re} s > 1/2$, by comparison with $N^{1/2-\sigma}$. Thus the Hardy limit is the displayed analytic function.

Uniform sum-integral comparison in $1/2 \leq \operatorname{Re} s \leq 3/4$, $|\operatorname{Im} s| \leq 2$, gives

$$P_N(s) = \frac{N^{1-s}}{1-s} + O(1).$$

Consequently, with $z = s - 1/2$, the function $G(z) = (1/2 - z)F(1/2 + z)$ equals $\sum_j a_j e^{-L_j z} + H(z)$, where H is uniformly bounded in the strip. Put $R = A^4$, $\Delta_k = L_{k+1} - L_k$. The two consecutive terms $B_k(z) = a_k e^{-L_k z} + a_{k+1} e^{-L_{k+1} z}$ have simple zeros

$$z_{k\ell} = \frac{\log A + (2\ell + 1)\pi i}{\Delta_k}.$$

On the circle $|z - z_{k\ell}| = 1/\Delta_k$, write $z = z_{k\ell} + u/\Delta_k$, $|u| = 1$. Then $|B_k(z)| = a_k |e^{-L_k z}||1 - e^{-u}| > a_k/8$, since $|1 - e^{-u}| \geq 3 - e > 1/4$ and $|e^{-L_k z}| > 1/2$.

Earlier blocks contribute at most $a_k/(A-1) = a_k/63$ on these circles; later blocks contribute less than $a_k/100$ by the supergeometric growth $L_{k+1} \geq R L_k$; the bounded error is less than $a_k/32$ for large k. Thus $|G - B_k| < |B_k|$ on each circle. Rouché's theorem gives exactly one zero of G in each disk. There are at least $\Delta_k/(4\pi)$ such disks with $|\operatorname{Im} z_{k\ell}| \leq 1/2$, and the bands for different k are disjoint. Each zero has $\operatorname{Re} z \geq (\log A - 1)/\Delta_k$ and $1 + |\rho|^2 < 3$, so the contribution from each band to the Blaschke sum is at least $(\log A - 1)/(12\pi)$. Summing over k diverges. □

AI assistance was used in adapting and checking the argument and the exposition.

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